MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW POWER SCHOTTKY TTL, REGISTERS, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Schottky TTL, low power, register microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit D-type register with three-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8") dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8") flat package
2	C-2 (20 terminal, .350" x .350") square chip carrier package
X	C-2A (20 terminal, .350" x .350") square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	
Device type 01	396 mW dc
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F, 2, and X	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.7 V dc
Case operating temperature range (T_C)	-55°C to +125°C
Minimum clock pulse width type 01	25 ns
Minimum clear pulse width type 01	25 ns
Minimum setup time at data enable type 01	35 ns
Minimum setup time at data type 01	17 ns
Minimum clear inactive state setup time type 01	10 ns
Minimum hold time at data enable type 01	0 ns
Minimum hold time at data type 01	7 ns
Maximum clock input frequency type 01	20 ns

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Truth table, terminal connections and logic diagrams. The truth table, terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V I_{OH} = -1 mA V_{IH} = 2.0 V V_{IL} = 0.7 V	01	2.4	---	V dc
Low level output voltage	V_{OL}	V_{CC} = 4.5 V I_{OL} = 12 mA V_{IL} = 0.7 V V_{IH} = 2.0 V	01	---	0.4	V dc
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V T_C = 25°C I_{IN} = -18 mA	01	---	-1.5	V dc
Low-level input current	I_{IL}	V_{CC} = 5.5 V V_{IN} = 0.4 V	01	-30	-380	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V V_{IN} = 2.7 V	01	---	20	µA
	I_{IH2}	V_{CC} = 5.5 V V_{IN} = 5.5 V	01	---	100	µA
Low-level output current outputs off	I_{OZL}	V_{CC} = 5.5 V V_O = 0.4 V V_I = 2.0 V	01	---	-20	µA
High-level output current outputs off	I_{OZH}	V_{CC} = 5.5 V V_O = 2.7 V V_I = 0.7 V	01	---	20	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	01	-30	-130	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 2/	01	---	30	mA
Maximum input clock frequency	f_{MAX}	V_{CC} = 5.0 V R_L = 110 Ω C_L = 50 pF	01	20	---	MHz
Propagation delay time high-to-low level clear to output	t_{PHL1}		01	7	46	ns
Propagation delay time low-to-high level clock to output	t_{PLH2}		01	7	33	ns
Propagation delay time high-to-low level clock to output	t_{PHL2}		01	7	39	ns
Propagation delay time disable to high level	t_{PZH}		01	2	34	ns
Propagation delay time disable to low level	t_{PZL}		01	2	35	ns
Propagation delay time high level to disable	t_{PHZ}		01	2	29	ns
Propagation delay time low level to disable	t_{PLZ}		01	2	29	ns

1/ Not more than one output should be shorted at a time.

2/ I_{CC} is measured with all outputs open; clear grounded following momentary connection to 4.5 V; N, G1, G2, and all data inputs grounded; and the clock input and M at 4.5 V.

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3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3, 7, 9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.3c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E using the circuit shown on figure 2, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E using the circuit shown on figure 2, or equivalent.

(2) T_A = 125°C minimum.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

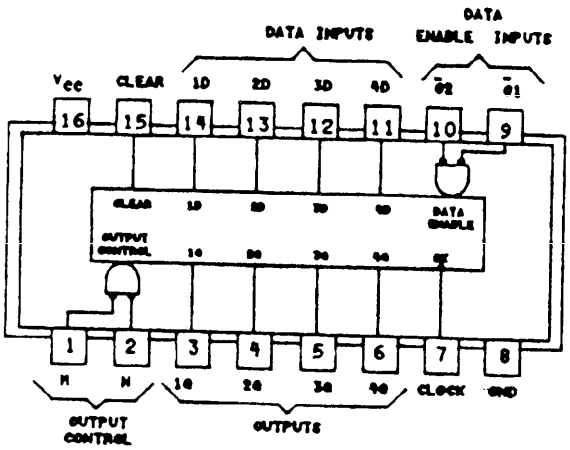
5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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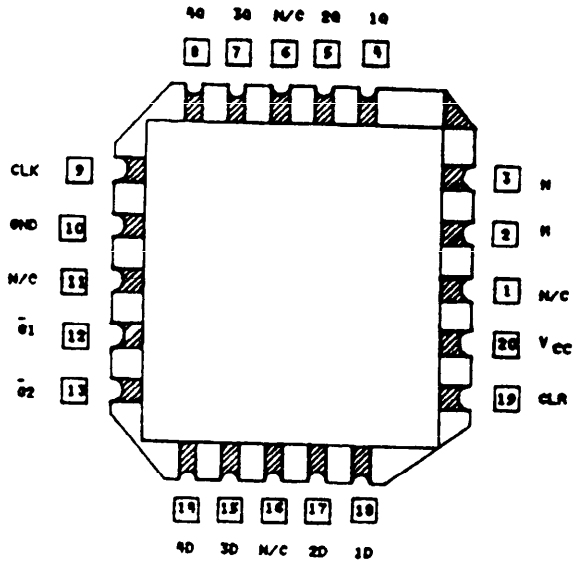
Device type 01

Cases E and F



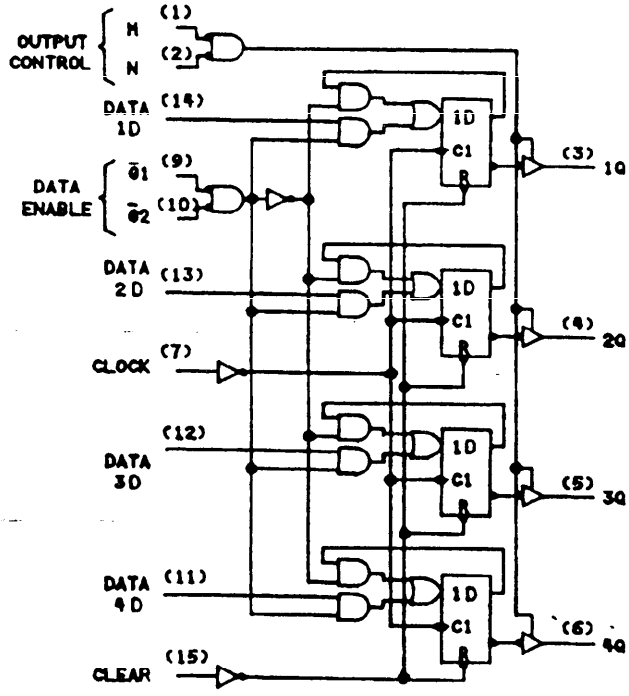
Device type 01

Case 2 X



Device type 01

Logic diagram



Device type 01

TRUTH TABLE

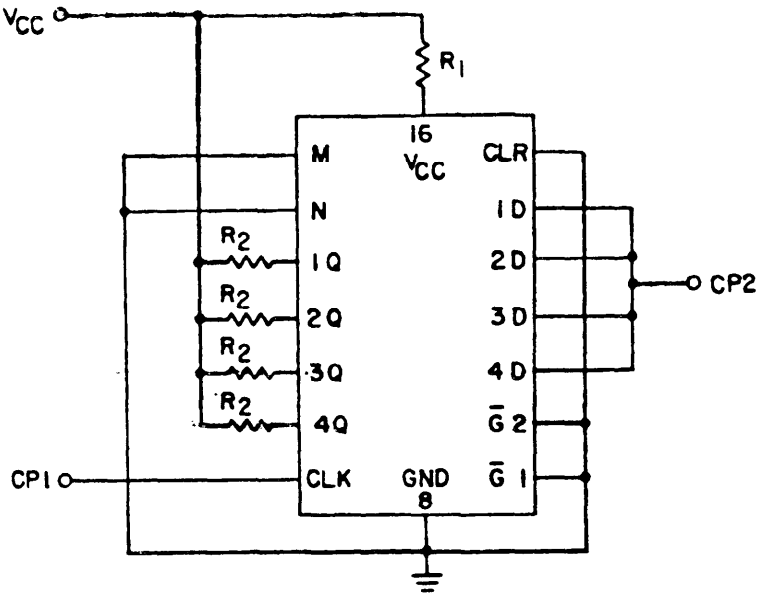
INPUTS					OUTPUT
CLEAR	CLOCK	DATA ENABLE $\bar{G}1$	DATA ENABLE $\bar{G}2$	DATA D	Q
H	X	X	X	X	L
L	L	X	X	X	Q_0
L	↑	H	X	X	Q_0
L	↑	X	H	X	Q_0
L	↑	L	L	L	L
L	↑	L	L	H	H

When either M or N (or both) is (are) high the output is disabled to the high-impedance state; however sequential operation of the flip-flops is not affected.

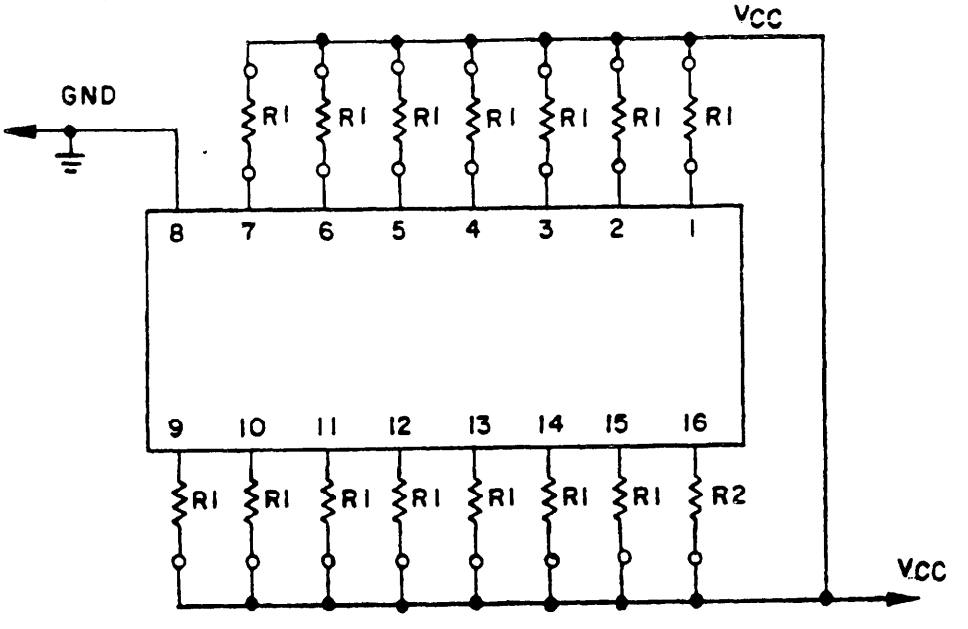
H = high level
L = low level
X = irrelevant

FIGURE 1. Terminal connections, truth table and logic diagram.

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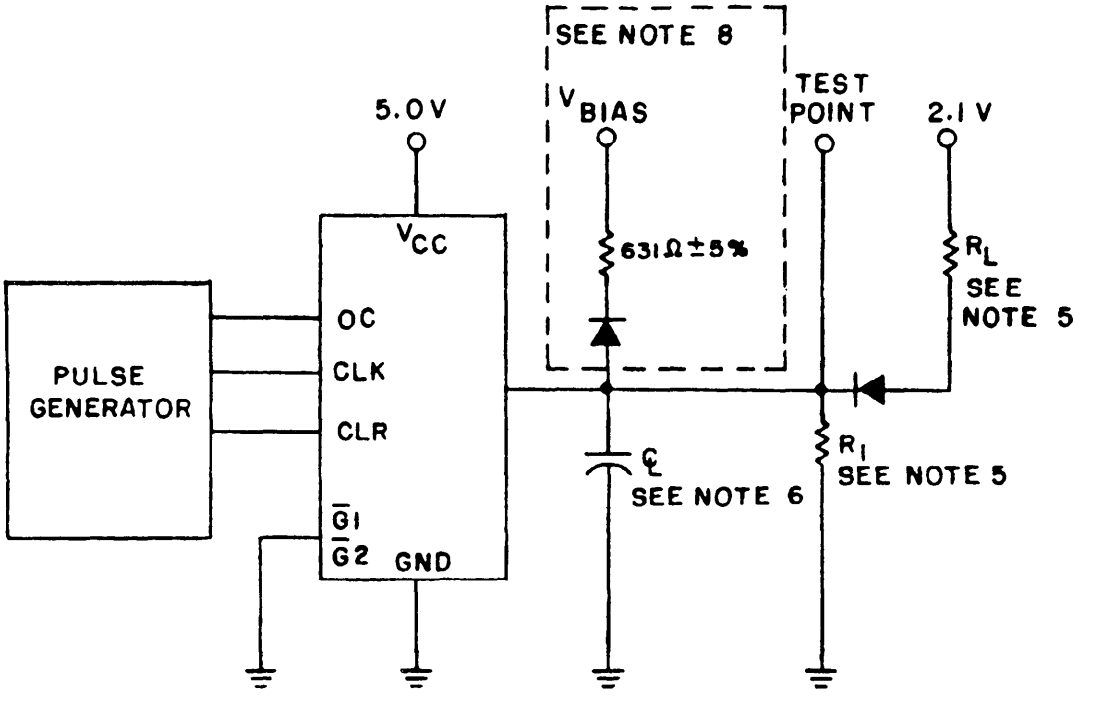
NOTES:
1. CP1 = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.
   CP2 = same as CP1, synchronized with CP1, except 50 kHz ±50 percent square wave.
2. R_2 = 470Ω ±5 percent.
3. V_{CC} and R_1 shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.



NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at V_{CC} terminal.

FIGURE 2. Burn-in and life test circuit for device type 01.

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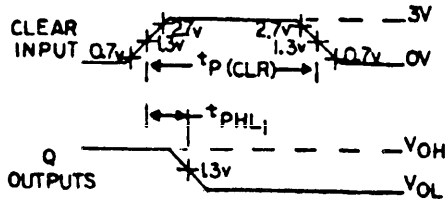


TEST CIRCUIT

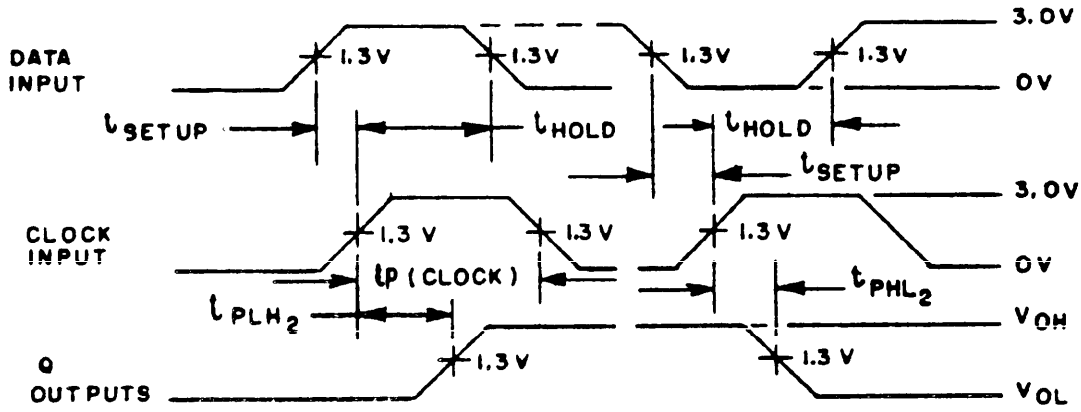
NOTES:
1. The pulse generator has the following characteristics; V_{GEN} = 3 V; PRR ≤ 1MHz; t_{TLH} ≤ 15 ns between 0.7 V and 2.7 V; t_{THL} ≤ 6 ns between 0.7 V and 2.7 V; Z_{OUT} = 50Ω.
2. Clear input pulse characteristics are as follows: $t_{p(CLR)}$ = 25 ns.
3. Clock input pulse characteristics are as follows: $t_{p(CLOCK)}$ = 25 ns.
4. Data input pulse characteristics are as follows: t_{SETUP} = 17 ns; t_{HOLD} = 7 ns.
5. R_1 = 2.4 kΩ ±5 percent; R_L = 110Ω ±5 percent.
6. C_L = 50 pF ±10 percent including scope probe, wiring and stray capacitance without package in test fixture.
7. All diodes are 1N3064 or equivalent.
8. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.

FIGURE 3. Switching time test circuit and waveforms.

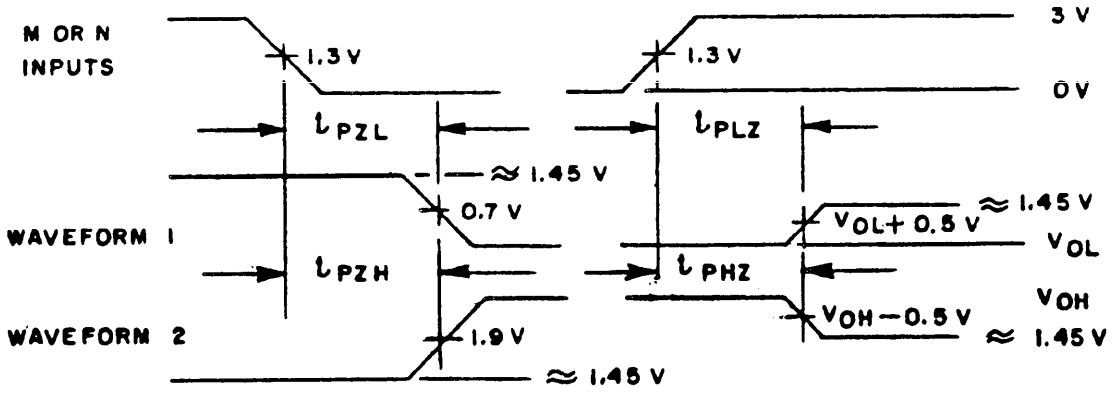
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CLEAR TO Q OUTPUTS



CLOCK TO Q OUTPUTS



TRI-STATE SWITCHING

FIGURE 4. Switching time waveforms.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 1/ 2, X / Test no.	1 / 2 / M	2 / 3 / N	3 / 4 / 1Q	4 / 5 / 2Q	5 / 7 / 3Q	6 / 8 / 4Q	7 / 9 / CLK	8 / 10 / GND	9 / 12 / $\overline{G1}$	10 / 13 / $\overline{G2}$	11 / 14 / 4D	12 / 15 / 3D	13 / 17 / 2D	14 / 18 / 1D	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1 2 3 4	0.7 V	0.7 V	-1 mA	-1 mA	-1 mA	-1 mA	2/	GND	0.7 V	0.7 V	2.0 V	2.0 V	2.0 V	2.0 V	0.7 V	4.5 V	1Q 2Q 3Q 4Q	2.4		V
	V_{OL}	3007	5 6 7 8			12 mA	12 mA	12 mA	12 mA					0.7 V	0.7 V	0.7 V	0.7 V			1Q 2Q 3Q 4Q		0.4	
	V_{IC}		9 10 11 12 13 14 15 16 17 18	-18 mA	-18 mA					-18 mA		-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA		M M CLK $\overline{G1}$ $\overline{G2}$ 4D 3D 2D 1D CLR		-1.5	
	I_{IL}	3009	19 20 21 22 23 24 25 26 27 28	0.4 V	0.4 V					0.4 V		0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	5.5 V	M M CLK $\overline{G1}$ $\overline{G2}$ 4D 3D 2D 1D CLR	3/	3/	μA
	I_{IH1}	3010	29 30 31 32 33 34 35 36 37 38	2.7 V	2.7 V					2.7 V		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		M M CLK $\overline{G1}$ $\overline{G2}$ 4D 3D 2D 1D CLR		20	
	I_{IH2}		39 40 41 42 43 44 45 46 47 48	5.5 V	5.5 V					5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		M M CLK $\overline{G1}$ $\overline{G2}$ 4D 3D 2D 1D CLR		100	
	I_{OZL}		49 50	2.0 V 2.0 V	2.0 V 2.0 V	0.4 V	0.4 V			2/ 2/						2.0 V	2.0 V	0.7 V 0.7 V		1Q 2Q		-20 -20	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 1/ 2, X / Test no.	1 / 2 / N	2 / 3 / N	3 / 4 / 1Q	4 / 5 / 2Q	5 / 7 / 3Q	6 / 8 / 4Q	7 / 9 / CLK	8 / 10 / GND	9 / 12 / $\overline{E1}$	10 / 13 / $\overline{E2}$	11 / 14 / 4D	12 / 15 / 3D	13 / 17 / 2D	14 / 18 / 1D	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{OZL}		51 52	2.0 V	2.0 V			0.4 V	0.4 V	2/	GND	GND	GND	2.0 V	2.0 V			0.7 V	5.5 V	3Q 4Q		-20 -20	µA
	I_{OZH}		53 54 55 56	"	"	2.7 V	2.7 V	2.7 V	2.7 V	"	"	"	"	0.7 V	0.7 V	0.7 V	0.7 V	"	"	1Q 2Q 3Q 4Q		20 " "	"
	I_{OS}	3011	57 58 59 60	GND	GND	GND	GND	GND	GND	"	"	"	"	5.5 V	5.5 V	5.5 V	5.5 V	GND	"	1Q 2Q 3Q 4Q	-30	-130	mA
	I_{CC}	3005	61	4.5 V	GND					4.5 V	"	"	"	GND	GND	GND	GND	4/	"	V_{CC}		30	"
2	Same tests, terminal conditions, and limits except T_C = +125°C and omit V_{IC} tests.																						
3	Same tests, terminal conditions, and limits except T_C = -55°C and omit V_{IC} tests.																						
7 T_C = +25°C	Truth table tests 5/	3014	62 63 64 65 66 67 68 69 70 71	B B B B B B B B B B	B B B B B B B B B B	L L H L L H L L H L	L L H L L H L L H L	L L H L L H L L H L	L L H L L H L L H L	B B A B B A B B A B	GND	B A B B A B B A B B	B A B B A B B A B B	B A A B A A B A A B	B A A B A A B A A B	B A A B A A B A A B	B A A B A A B A A B	A B B A B B A B B A	4.5 V	6/			
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 7/	3003 (Fig 4)	72 73 74 75	GND	GND	OUT	OUT	OUT	OUT	IN	GND	GND	GND	IN	IN	IN	IN	GND	5.0 V	1Q 2Q 3Q 4Q	20		MHz
	t_{PHL1}		76 77 78 79	"	"	OUT	OUT	OUT	OUT	2/	"	"	"	4.5 V	4.5 V	4.5 V	4.5 V	IN	"	CLR to 1Q CLR to 2Q CLR to 3Q CLR to 4Q	7	35	ns
	t_{PLH2}		80 81 82 83	"	"	OUT	OUT	OUT	OUT	IN	"	"	"	IN	IN	IN	IN	4/	"	CLK to 1Q CLK to 2Q CLK to 3Q CLK to 4Q	"	25	"
	t_{PHL2}		84 85 86 87	"	"	OUT	OUT	OUT	OUT	"	"	"	"	IN	IN	IN	IN	GND	"	CLK to 1Q CLK to 2Q CLK to 3Q CLK to 4Q	"	30	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 1/ 2, X / Test no.	1 / 2 / M	2 / 3 / M	3 / 4 / 1Q	4 / 5 / 2Q	5 / 7 / 3Q	6 / 8 / 4Q	7 / 9 / CLK	8 / 10 / GND	9 / 12 / $\overline{G1}$	10 / 13 / $\overline{G2}$	11 / 14 / 4D	12 / 15 / 3D	13 / 17 / 2D	14 / 18 / 1D	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PZH}	3003 (fig 4)	88 89 90 91 92 93 94 95	IN " " " GND " " "	GND " " " IN " " "	OUT " " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	2/ " " " " " " "	GND " " " " " " "	GND " " " " " " "	GND " " " " " " "	4.5 V " " 4.5 V " " " 4.5 V	4.5 V " " " " " " 4.5 V	4.5 V " " " 4.5 V " " "	4.5 V " " " 4.5 V " " "	GND " " " " " " "	5.0 V " " " " " " "	M to 1Q M to 2Q M to 3Q M to 4Q M to 1Q M to 2Q M to 3Q M to 4Q	2 " " " " " " "	26 " " " " " " "	ns " " " " " " "
	t_{PZL}		96 97 98 99 100 101 102 103	IN " " " GND " " "	GND " " " IN " " "	OUT " " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	GND " " " GND " " "	GND " " " " " " "	GND " " " GND " " "	GND " " " GND " " "	" " " " " " " "	" " " " " " " "	M to 1Q M to 2Q M to 3Q M to 4Q M to 1Q M to 2Q M to 3Q M to 4Q	" " " " " " " "	27 " " " " " " "	" " " " " " " "
	t_{PHZ}		104 105 106 107 108 109 110 111	IN " " " GND " " "	GND " " " IN " " "	OUT " " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	4.5 V " " 4.5 V " " " 4.5 V	4.5 V " " " " " " 4.5 V	4.5 V " " " 4.5 V " " "	4.5 V " " " 4.5 V " " "	" " " " " " " "	" " " " " " " "	M to 1Q M to 2Q M to 3Q M to 4Q M to 1Q M to 2Q M to 3Q M to 4Q	" " " " " " " "	22 " " " " " " "	" " " " " " " "
	t_{PLZ}		112 113 114 115 116 117 118 119	IN " " " GND " " "	GND " " " IN " " "	OUT " " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	OUT " " OUT " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	" " " " " " " "	GND " " " GND " " "	GND " " " " " " "	GND " " " GND " " "	GND " " " GND " " "	" " " " " " " "	" " " " " " " "	M to 1Q M to 2Q M to 3Q M to 4Q M to 1Q M to 2Q M to 3Q M to 4Q	" " " " " " " "	" " " " " " " "	" " " " " " " "

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 1/ 2, X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	M	N	1Q	2Q	3Q	4Q	CLK	GND	$\overline{G}1$	$\overline{G}2$	4D	3D	2D	1D	CLR	V_{CC}				
10 T_C = +125°C	f_{MAX}																				20		MHz
	t_{PHL1}																				7	46	ns
	t_{PLH2}																				"	33	"
	t_{PHL2}																				"	39	"
	t_{PZH}																				2	34	"
	t_{PZL}																				"	35	"
	t_{PHZ}																				"	29	"
	t_{PLZ}																				"	29	"
			Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																				
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

Notes:

1/ Pins not referenced are N/C.

2/ Apply one clock pulse prior to test; 2.5 V minimum to 5.5 V maximum, 0 V

3/ I_{IL} min/max values (µA) for circuits shown:

Measured terminal	A	B	C
M, N, $\overline{G}1$, $\overline{G}2$, CLR	-120/-360	-120/-380	-120/-380
4D, 3D, 2D, 1D	-120/-360	-150/-380	-120/-380
CLK			-30/-300

4/ Apply one clear pulse prior to test; 4.5 V, 0 V

5/ A = 3.0 V minimum; B = 0.0 V or GND.

6/ H > 1.5 V; L < 1.5 V.

7/ The f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half the clock input frequency. The D input shall be one-half of the frequency of the clock and the D ↑ and ↓ shall be nominally coincident with the clock ↓ but may be offset sufficiently to assure adequate t_{SETUP} and t_{HOLD} (see 1.4).

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal.
V_{IN} - - - - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS173A

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6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	A Texas Instruments	B Motorola Inc.	C Signetics
01	X	X	X

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1100)

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